EE6301
Concept Framework
中英文知识框架

8 weeks · 64 framework entries
8周 · 64条知识联系 · 2026-10-13

Body text / 正文：11 pt
English first, Chinese immediately below. / 英文在前，中文紧随。

2026-10-13 · Course notes, sample explanations and original practice
课件知识、样题参考解析与原创练习

Contents / 内容目录

Week 1: Biosensors and measurement quality / 第1周：生物传感器与测量质量 3

Week 2: Light and optical signals / 第2周：光与光学信号 4

Week 3: Labels, fibres and resonance / 第3周：标记、光纤与共振 5

Week 4: Microscopes and optical imaging / 第4周：显微镜与光学成像 6

Week 5: Ultrasound, photoacoustics and image analysis / 第5周：超声、光声与图像分析 7

Week 6: Electrochemical biosensors / 第6周：电化学生物传感器 8

Week 7: Bioelectricity, ECG and EEG / 第7周：生物电、心电与脑电 9

Week 8: Cell recording, amplification and filters / 第8周：细胞记录、放大与滤波 11

EN: Use these maps to establish the relationships between topics. The full learning guide supplies explanations, examples, conditions, figures and the reviewed 25-question mock. Follow P1/P2/P3 priorities in the full guide.

中文： 用框架建立知识之间的关系；完整版提供讲解、例题、适用条件、配图和已复核的25题模拟。复习优先级见完整版的P1/P2/P3。

Week 1: Biosensors and measurement quality / 第1周：生物传感器与测量质量

Week 1 concept map / 第1周知识框架

EN: Chapter logic: A biosensor recognizes a target, converts its event into a signal, and uses calibration to judge identity and amount.

中文： 本章主线：生物传感器先识别目标，再把事件变成信号，最后用校准判断是什么、多少及能否检出。

English term 中文名词	Meaning and connection 含义与知识联系
Analyte • bioreceptor • transducer 分析物•生物受体•换能器	Target, recognition and signal conversion are separate roles. 被测目标、识别与信号转换是三个不同角色。 → C01
Biorecognition 生物识别	Antibodies bind; enzymes catalyse; complementary nucleic acids hybridize. 抗体结合，酶催化，互补核酸杂交。 → B01
Optical • electrochemical • mechanical 光学•电化学•机械	Classify the transducer by the measured physical change. 按测到的物理变化判断换能类别。 → B01
Sensitivity • selectivity 灵敏度•选择性	Sensitivity is calibration slope; selectivity concerns interferents. 灵敏度是校准斜率，选择性针对干扰物。 → C02
Limit of detection, LOD 检测限	LOD=3σ/|m| combines blank noise with linear calibration slope. LOD=3σ/|m|同时考虑空白噪声与线性校准斜率。 → C02
Linearity • response time • reproducibility 线性度•响应时间•再现性	These describe range, speed and consistency; they do not alone establish accuracy. 分别描述范围、速度和一致性，不能单独证明准确。 → B02
POCT • microfluidics 即时检测•微流控	POCT describes test location; microfluidics organizes small-volume sample handling. POCT描述检测地点，微流控组织少量液体处理。 → B03
Minimum integer count 最小整数数量	Round upward only when a minimum integer count must reach a continuous threshold. 仅当最小整数数量必须达到连续阈值时，向上取整。 → C03

Week 2: Light and optical signals / 第2周：光与光学信号

Week 2 concept map / 第2周知识框架

EN: Chapter logic: Match a light source to the sample, follow how light interacts with it, and distinguish the measured optical quantities.

中文： 本章主线：先让光源匹配样品，再追踪光怎样作用，最后区分实际测得的光学量。

English term 中文名词	Meaning and connection 含义与知识联系
Ray • wave • photon 光线•波•光子	Paths, interference and energy exchange call for different light models. 路径、干涉与能量交换使用不同光模型。 → B04
Photon energy • photon rate 光子能量•光子速率	$E=hc/\lambda$ describes one photon; P/E gives photons per second. $E=hc/\lambda$描述一个光子，P/E得到每秒光子数。 → S105
Coherence • noise • power 相干性•噪声•功率	Select suitable phase stability, noise and power; brighter is not automatically better. 相位稳定、噪声和功率须合适，越亮不一定越好。 → S104
Absorption • scattering • attenuation 吸收•散射•衰减	Direct-beam attenuation can contain both absorption and redirection losses. 直达光衰减可同时包含吸收与改变方向造成的损失。 → B05
Absorbance • transmittance 吸光度•透射率	$A=-\log_{10}T=\varepsilon CL$; compare at the same wavelength and matching units. $A=-\log_{10}T=\varepsilon CL$；同波长比较，并匹配单位。 → C04
Pulse oximetry 脉搏血氧测量	Two wavelengths and the pulsatile component help estimate arterial saturation. 两波长与脉动分量帮助估计动脉血氧饱和度。 → B06
Fluorescence • filters • Φ • τ 荧光•滤光片•产率•寿命	Match excitation and emission bands; Φ is emitted/absorbed and $k_r=\Phi/\tau$. 匹配激发/发射波段；Φ=发射/吸收，$k_r=\Phi/\tau$。 → C05
Raman • Rayleigh • spectral resolution 拉曼•瑞利•光谱分辨	Raman exchanges vibration energy; Rayleigh is elastic; grating $R=mN$ separates wavelengths. 拉曼交换振动能，瑞利为弹性；光栅$R=mN$描述分开波长的能力。 → C07

Week 3: Labels, fibres and resonance / 第3周：标记、光纤与共振

Week 3 concept map / 第3周知识框架

EN: Chapter logic: Follow how unlabelled binding changes a guided or resonant optical response, while separating recognition from the readout.

中文： 本章主线：追踪无报告标签的结合怎样改变导光或共振响应，并始终区分识别与读出。

English term 中文名词	Meaning and connection 含义与知识联系
Labelled • label-free 有标记•无标记	An added reporter defines labelled sensing; label-free does not mean no light or no receptor. 添加报告物是有标记；无标记不等于不用光或不用受体。 → C09
Quantum dots • flow cytometry 量子点•流式细胞术	Dot size can tune color within a material family; flow cytometry reads individual passing events. 同材料系列量子点尺寸可调颜色；流式逐个读取经过的事件。 → B09
Total internal reflection • fibre NA 全内反射•光纤NA	TIR needs high-to-low index and an angle above critical; acceptance and internal angles differ. TIR要求高到低折射率及超过临界角；接受角不同于内部角。 → S114
Evanescent field 倏逝场	A rapidly decaying external field lets nearby molecules affect a guided optical signal. 外侧迅速衰减的场使附近分子影响导光信号。 → S115
Modes • sensing arrangement 模式•传感布置	Mode count is a field property; point, multiplexed and distributed describe measurement positions. 模式数是场性质；点式、复用、分布式描述测量位置。 → S116
Optical resonators • SPR • LSPR 光学共振器•SPR•LSPR	Identify circulating/standing optical patterns versus metal collective-electron resonance. 辨认循环/驻波光场与金属集体电子共振。 → C11
Sλ • FWHM • Q • FOM 位移灵敏度•线宽•Q•FOM	Compare shift with width in consistent units; index sensitivity alone does not establish selectivity. 位移和线宽要统一单位；折射率灵敏度本身不证明选择性。 → B11
ART • SERS 抗原快检•表面增强拉曼	ART uses antibody capture and gold-particle color; SERS enhances molecular vibration fingerprints. ART以抗体捕获并用金颗粒显色；SERS增强分子振动指纹。 → C12

Week 4: Microscopes and optical imaging / 第4周：显微镜与光学成像

Week 4 concept map / 第4周知识框架

EN: Chapter logic: Follow light from image formation and contrast to depth selection, detection and spectral information.

中文： 本章主线：沿着光的流程，理解成像与对比如何产生，再看怎样选深度、检测光和获取光谱信息。

English term 中文名词	Meaning and connection 含义与知识联系
Imaging modalities 成像模态	Identify the input energy and detected signal; projection X-ray and reconstructed CT slices differ. 先辨输入能量与检测信号，区分投影X射线与重建CT切片。 → B13
NA, resolution and sampling NA、分辨率与采样	d≈0.61λ/NA: larger NA or shorter λ improves lateral separation; pixels must sample the magnified detail. d≈0.61λ/NA：大NA或短λ改善横向间距；像素须充分采样放大后的细节。 → C13
Optical contrast 光学对比	Brightfield, darkfield, phase contrast and fluorescence use different sources of image contrast. 明场、暗场、相衬与荧光使用不同对比来源。 → S402
Confocal versus two-photon 共聚焦与双光子	Confocal rejects light at detection; two-photon localizes excitation ($F \propto I^2$ before saturation). 共聚焦在检测端拒光；双光子限制激发位置（未饱和时$F \propto I^2$）。 → C14
OCT mechanism OCT机制	Sample return + reference light → interference → depth-reflectivity information. 样品返回光+参考光→干涉→深度反射信息。 → C15
Three depth concepts 三种深度概念	OCT axial separation follows source bandwidth; lateral detail follows focusing; focused range is another quantity. OCT轴向间距由光源带宽控制，横向细节依赖聚焦，可聚焦范围是另一量。 → B17
Optical detectors 光探测器	Photodiode/PMT convert light to current; CCD shifts charge, while CMOS uses pixel-associated circuits. 光电二极管/PMT把光转电流；CCD移电荷，CMOS用像素相关电路。 → C16
Hyperspectral cube 高光谱立方	Each position has a spectrum: axes are x,y,λ; scan method determines what is acquired per step. 每个位置有光谱，坐标是x,y,λ；扫描方式决定一步记录什么。 → S406

Week 5: Ultrasound, photoacoustics and image analysis / 第5周：超声、光声与图像分析

Week 5 concept map / 第5周知识框架

EN: Chapter logic: Compare what enters tissue and what returns, then turn measured signals into images and diagnostic decisions.

中文： 本章主线：比较送入组织和返回的信号，再理解测量怎样形成图像及诊断判断。

English term 中文名词	Meaning and connection 含义与知识联系
Piezoelectric transducer 压电换能器	Voltage→deformation transmits sound; pressure→charge receives it. 电压→形变用于发声，压力→电荷用于接收。 → B18
Acoustic impedance and reflection 声阻抗与反射	$Z=\rho c$; impedance mismatch creates echoes, and high frequency usually trades depth for detail. $Z=\rho c$；失配形成回波，高频通常以穿透深度换更细细节。 → S501
Echo versus photoacoustic timing 回波与光声计时	Echo depth uses ct/2 for the round trip; a stated one-way photoacoustic route uses ct. 回波按往返ct/2，明确单程的光声声路按ct。 → C17
Photoacoustic contrast 光声对比	Absorbed pulsed light→heating/expansion→sound; multiple wavelengths can reveal different absorbers. 脉冲光吸收→升温/膨胀→声音，多波长可区分不同吸光物。 → S503
X-ray and CT X射线与CT	$I/I_0=\exp(-\Sigma\mu_i L_i)$ for stated uniform layers; CT combines angular projections into slices. 题设均匀多层$I/I_0=\exp(-\Sigma\mu_i L_i)$，CT合并角度投影重建切片。 → S504
Digital-image representation 数字图像表达	Sampling discretizes position; b-bit quantization gives 2^b intensity levels. 采样离散化位置，b位量化给2^b个强度等级。 → B20
Processing and analysis 处理与分析	Clean the image, segment regions, measure features, then classify, estimate values or track objects. 整理图像、分割区域、测特征，再分类、估数值或跟踪。 → S505
Diagnostic metrics 诊断指标	TP/TN/FP/FN compare predictions with truth; ROC plots TPR against FPR as threshold changes. TP/TN/FP/FN对比预测和真值；ROC在变阈值时画TPR对FPR。 → S505

Week 6: Electrochemical biosensors / 第6周：电化学生物传感器

Week 6 concept map / 第6周知识框架

EN: Chapter logic: Link chemical recognition or molecular charge to the electrical variable measured by the sensor.

中文： 本章主线：把化学识别或分子电荷，与传感器实际测得的电量连起来。

English term 中文名词	Meaning and connection 含义与知识联系
Redox and reference 氧化还原与参比	Oxidation loses electrons, reduction gains them; electrode potential is relative to a stated reference. 氧化失电子、还原得电子；电极电位相对给定参比。 → B21
Potentiometric/ion-selective sensing 电位型/离子选择感测	Read potential at negligible current; an ideal Nernst response is logarithmic in ion activity. 在极小测量电流下读电位，理想Nernst响应与离子活度呈对数关系。 → S603
Amperometry and glucose 安培型与葡萄糖	Controlled potential measures reaction current; distinguish enzyme recognition from electrode readout and GOx generations. 控制电位测反应电流，区分酶识别、电极读出与GOx各代路线。 → C19
Charge and stoichiometry 电荷与反应计量	For constant current Q=It; with complete charge collection $Q=z\,n_a F$. 恒定电流Q=It；完全收集电荷时$Q=z\,n_a F$。 → B24
Conductance and impedance 电导与阻抗	G=1/R; AC impedance adds magnitude, phase and frequency dependence. G=1/R；交流阻抗还包含幅值、相位及频率依赖。 → B26
FET, nanopore and capacitance FET、纳米孔与电容	Gate charge changes a channel; pore passage changes ionic current; dielectric/geometry changes capacitance. 栅电荷改变沟道，穿孔改变离子电流，介电/几何变化改变电容。 → B25
Molecular charge 分子电荷	DNA is typically negative; protein net charge varies with pH/pI, so bound charge is not universally positive or negative. DNA通常负电，蛋白净电荷随pH/pI变，因此结合电荷并非普遍同正或同负。 → S605
Device fabrication 器件制造	Photolithography patterns resist; etching removes, deposition adds, and PDMS molding copies microstructures. 光刻形成胶图案，刻蚀去材料、沉积加材料，PDMS模制复制微结构。 → B22

Week 7: Bioelectricity, ECG and EEG / 第7周：生物电、心电与脑电

Week 7 concept map / 第7周知识框架

EN: Chapter logic: Ionic gradients and selective channels create cell voltages; tissue activity reaches electrodes as potential differences, which are interpreted as EMG, ECG or EEG patterns.

中文： 本章主线：离子梯度与选择性通道形成细胞电压；组织活动以电位差到达电极，再根据来源与时间模式解释为EMG、ECG或EEG。

English term 中文名词	Meaning and connection 含义与知识联系
Bioelectricity / biopotential 生物电 / 生物电位	Recording measures potential differences; stimulation deliberately applies an electrical input. ECG, EMG and EEG identify heart, muscle and brain-related recordings. 记录测电位差；刺激主动施加电输入。ECG、EMG、EEG分别对应心脏、肌肉及脑相关记录。 → C20
Membrane potential and AP phases 膜电位与动作电位阶段	Vm=inside−outside. Na^+ entry raises the taught neuronal voltage; later K^+ exit lowers it, with a possible dip below rest. Vm=胞内−胞外。课程神经元中Na^+内流使电压升高，随后K^+外流使其降低，可短暂低于静息。 → C21
Threshold / all-or-none / firing rate 阈值 / 全或无 / 发放频率	Threshold triggers a spike; stronger input commonly produces more similar spikes per second, not proportionally taller spikes. 阈值决定是否触发；更强输入通常增加每秒相似尖峰数量，不是让单个尖峰成比例变高。 → B27
Nernst versus GHK Nernst与GHK	Nernst balances one ion; GHK combines ions with different permeabilities. Zero summed current does not require every ionic current to vanish. Nernst平衡一种离子，GHK综合不同通透性的多种离子。电流总和为零不要求每项都为零。 → B28
Electrode–electrolyte interface 电极—电解质界面	Ionic tissue current meets electronic metal conduction; the interface adds voltage offsets and frequency-dependent impedance. 组织离子载流与金属电子载流在界面交接；界面加入电压偏移及随频率变化的阻抗。 → B29
P / QRS / T P波 / QRS波群 / T波	P: atrial depolarization; QRS: ventricular depolarization; T: ventricular repolarization. These are waveform events, not separate leads. P：心房去极化；QRS：心室去极化；T：心室复极化。这是曲线事件，不是三个导联。 → C22

English term 中文名词	Meaning and connection 含义与知识联系
ECG leads and RR interval 心电导联与RR间期	A lead defines a voltage-difference view; II=I+III for the given limb definitions. With RR in seconds, heart rate=60/RR. 导联定义一种电位差视角；所给肢体定义满足II=I+III。RR用秒时，心率=60/RR。 → B30
Scalp EEG and wave frequency 头皮脑电与脑波频率	EEG reflects combined neuronal activity at standardized sites. Frequency comes from complete cycles/elapsed seconds; amplitude is a different quantity. EEG在标准位置记录神经元共同活动。频率=完整周期数/实际秒数；幅度是另一个量。 → B31

Week 8: Cell recording, amplification and filters / 第8周：细胞记录、放大与滤波

Week 8 concept map / 第8周知识框架

EN: Chapter logic: Choose a cell-recording method, preserve and amplify the weak signal, then select its useful frequency band; this chapter stops at Parts One and Two through PDF p29.

中文： 本章主线：先选择细胞记录方法，再保住并放大弱信号，最后选择有用频段；本组只到原PDF p29的第一、二章。

English term 中文名词	Meaning and connection 含义与知识联系
Patch clamp 膜片钳	A pipette forms a tight membrane seal. Voltage clamp holds voltage/read current; current clamp applies current/read voltage. 吸管与膜紧密封接。电压钳保持电压、读电流；电流钳施加电流、读电压。 → C23
MEA / raster plot 多电极阵列 / 栅格图	MEA records extracellular activity from many sites in parallel. A raster plot shows when each channel detects spikes. MEA并行记录多个位点的细胞外活动；栅格图显示各通道何时检测到尖峰。 → S801
Differential / common-mode 差分 / 共模	Vout=G(V1−V2). Equal shared interference cancels ideally; unequal interference remains. Vout=G(V1−V2)。相同共有干扰理想抵消，不同干扰仍可留下。 → C24
Input impedance / output impedance / gain 输入阻抗 / 输出阻抗 / 增益	High input impedance reduces loading; low output impedance drives a load; gain specifies amplification. They are independent roles. 高输入阻抗减少加载，低输出阻抗驱动负载，增益决定放大倍数；三者分工不同。 → S802
Preamplifier / isolation 前置放大 / 隔离	The preamplifier handles a weak input; isolation limits unwanted electrical paths and supports patient protection. 前置放大处理弱输入；隔离限制不希望出现的电连接路径，帮助保护被测者。 → B32
Inverting / non-inverting 反相 / 同相	In the shown ideal circuits, gains are −Rf/Ri and 1+Rf/Ri. The minus sign reverses polarity, not amplification itself. 所示理想电路增益为−Rf/Ri与1+Rf/Ri。负号表示极性翻转，不表示没有放大。 → S803

English term 中文名词	Meaning and connection 含义与知识联系
Low-pass / high-pass / band-pass / notch 低通 / 高通 / 带通 / 陷波	Filters attenuate unwanted frequency components: low-pass keeps slower components, high-pass faster, band-pass an interval, notch a narrow rejected band. 滤波器削弱不需要的频率成分：低通保留较慢、高通较快、带通保留区间、陷波抑制窄频带。 → C25
Frequency-response graph 频率响应图	Horizontal frequency and vertical gain reveal the retained region. The local RC worked example is excluded; basic filter-role recognition remains. 横轴频率、纵轴增益显示保留区域。局部RC例题被排除，基本滤波功能辨认仍保留。 → S804